

Heart Disease Risk Assessment using Agentic AI

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ABSTRACT — Heart diseases remain some of the most frequent causes of mortality in the world, and being able to detect them at the early stages may help many people prevent them. In this paper, I am proposing a system for predicting the likelihood of having a heart disease using a machine learning algorithm coupled with an agentic approach to artificial intelligence and implemented into a web application. The predictor takes information about the patient such as age, gender, type of chest pain, resting blood pressure, cholesterol level, fasting blood sugar, electrocardiographic (ECG) measurements, maximum heart rate achieved during stress test, exercise induced angina, ST depression measurement, and the slope of the ST segment. Out of the tested models, the Random Forest classifier with 300 estimators and class weights = "balanced" was selected as the most appropriate one due to its ability to deal with mixed variables, extract non-linear relationships between features and avoid overfitting by default. The backend involves using the Flask REST API where the application runs and which provides specific endpoints for prediction, follow-up chat, history, and PDF reporting. As for the frontend, it has been completely custom-built with HTML, CSS, and JS technologies, with the dark glass morphism theme applied to it and real-time 3D particle animation used as an extra layer of visual effects. One of the most interesting aspects of this project is multi-agency — that is, there are different AI agents, all running on the Groq cloud API with large language models, designed to convert patient descriptions to structured form, provide medical reasoning in simple language, suggest individual lifestyle modifications, conduct follow-ups, create PDF reports for download, and store historical data on predictions made so far. Performance-wise, the Random Forest model scored 89.67% accuracy, 88.79% precision, 93.14% recall, 90.91% F1-score, and 93.50% ROC AUC. In conclusion,

the resulting application is not limited to just being a predictor, which provides its users with a concrete risk assessment score, an explanation of the factors influencing the risk assessment, practical recommendations on health issues, a chat where any user-related questions can be asked, and finally, the option to download the created PDF report. Thus, one can see how a synergy between machine learning methods and the use of agents can lead to a helpful health risk assessment tool.

Keywords: Heart disease prediction, Random Forest, machine learning, agentic AI, Flask REST API, single-page application, healthcare analytics, clinical decision support.

1. INTRODUCTION

The threat of heart diseases remains to be one of the leading factors responsible for fatalities and disabilities all around the world, taking lives of numerous patients yearly regardless of their age group or geographic location. One of the biggest problems associated with this pathology is that many people fail to comprehend the fact that they suffer from the disorder until it gets into its critical stage. Therefore, early diagnostics and preventive awareness play a vital role when trying to tackle the problem. Up until recently, checking a person's cardiovascular condition could only be done in a conventional way via medical tests, analyses, and the assessment of physicians' experience. Although being still considered the most accurate way, such practices were not available to all, especially those living in areas of inadequate healthcare services. Machine learning technologies allow for estimating the risks of heart diseases for a patient by analyzing patterns inherent in his/her data and providing an instant estimation prior to any medical symptoms development. Thus, the objective of this project is not just to design a classifier

distinguishing between "risk" and "no risk." It aims at developing an all-around system for the patient. It computes the numeric probability of a risk factor, provides detailed explanations in English for the reason why the given risk factor was identified at a certain level, provides personalized advice on lifestyle and diet changes, enables users to interact with the program in case they have any queries via a chatbot feature driven by artificial intelligence technology, saves the prediction record of every query for later reference, and creates a professional PDF report which can be presented to the healthcare provider. For implementing the above functionality, the project integrates a random forest machine learning algorithm with a multi-agent AI pipeline, in which each individual agent plays a particular role in the workflow. It is implemented as a state-of-the-art web-based application, with its backend built as a Flask REST API to perform the task of predictions, chats, reports, and histories through well-defined API endpoints, and the frontend as a customized single-page application designed using HTML, CSS, and JavaScript to render an engaging user interface.

2. PROBLEM STATEMENT

The key issue that the proposed study attempts to address lies in the lack of an easily available and intelligent solution which can assess the risk of heart diseases based on a simple clinical input and deliver intelligent advice beyond the mere risk assessment. Merely delivering a risk score such as "high risk" or "low risk" does not suffice; the user needs to know the reasoning behind this classification and also what measures can be taken for reducing risks.

The goals of this project include:

- (1) Preprocessing structured heart disease data, taking into account numerical and categorical data for machine learning.
- (2) Training and testing a Random Forest classifier to predict the risk of heart disease.
- (3) Providing a risk probability along with the class prediction in order to make the result more understandable for the user.
- (4) Generating medical reasoning with the use of large language models by leveraging the Groq API, in clear language understandable to non-experts.
- (5) Making recommendations for a personalized diet, exercise, and healthy lifestyle changes that will be recommended to improve the user's condition.
- (6) Offering a live chat opportunity where users can ask any question regarding the prediction result.
- (7) Generating a PDF report that users can download as well as saving history of previous predictions.

3. LITERATURE REVIEW

"Cardiovascular diseases (CVDs)" [1] is an overview of the huge mortality burden of CVDs, their classification based on pathophysiology, and risk factors related to these diseases. It is pointed out that in order to control this major killer disease, which is responsible for causing deaths across the world, there should be public health approaches focusing on behavioral, environmental, and health system risk factors.

"Heart disease prediction comparison through logistic regression, SVM, KNN, and random forest using cross-validation for enhanced accuracy" [2] focuses on the efficiency of four approaches within machine learning aimed at recognizing symptoms of cardiovascular diseases. In accordance with the initial article, the random forest model was more efficient compared to other three models due to the greater accuracy (F1 score – 95%) in comparison to logistic regression (accuracy – 84%). Nevertheless, it is worth noting that the above-referenced scientific article has been officially retracted from the Scientific Reports due to language misuse.

"Visualizing the Whole Spectrum Optimization of K-Nearest Neighbors from Data Pre-processing to Hyperparameter Optimization and K-Fold Cross-Validation for Predicting Cardiovascular Disease" [3] explains a complete machine learning system developed specifically to increase the effectiveness of the K-Nearest Neighbors (KNN) method in predicting heart disease early on. This study stresses the necessity of comprehensive model optimization by using the Hampel filter to eliminate any outliers present, applying PCA as a feature selection technique, and conducting grid search alongside 10-folds cross-validation in optimizing the hyperparameters. It is proven that this optimization process greatly improves the predictive performance of the KNN model, making it possible to predict with 92.46% accuracy from the Framingham dataset.

"A comparative study of machine learning methods for heart stroke prediction" [4] compares nine machine learning methods such as SVM, KNN, XGBoost, and Random Forest for the effective prediction and classification of stroke diseases using known risk factors. According to the study, the model based on the random forest algorithm performed better than all other tested algorithms and had an accuracy rate of 98.4%.

"GUI Base Prediction of Heart Stroke Stages by Using the Accuracy of Machine Learning Algorithm" [5] outlines the design of an algorithm capable of using the powers of machine learning and supervised classification in determining the severity and stages of heart strokes. In addition, the research presents an easy-to-use Graphical User Interface (GUI) to be used in the classification and

prediction of the stroke risks using various factors, including the symptoms of atrial fibrillation. According to the study, the integration of machine learning in this analysis yields very impressive results, particularly in terms of sensitivity at around 99.98%.

"AMIE gets eyes: A research AI agent for multimodal diagnostic dialogue" [6] discusses how a vision-enabled AI agent can be developed in order to incorporate both visual data and textual conversations in order to facilitate better clinical reasoning. This agent actively learns from the conversation and strategically asks for and analyzes different types of medical images, including those used for dermatology and x-rays, in the context of a conversation in order to fill knowledge gaps. In the experimental setup of clinical dialogues, this vision-enabled AI was able to perform at least as well and often significantly better than board-certified doctors for general practitioners when it came to key metrics.

"Performance Analysis of Machine Learning Techniques for Stroke Prediction" [7] investigates different machine learning classifiers in predicting the chances of stroke based on certain health parameters, such as blood pressure, body mass index, cardiovascular disease, and average blood sugar level. This study reveals that through the integration of different models, the suggested weighted voting classifier surpassed all other individual conventional algorithms with an impressive prediction accuracy rate of 97%.

"Heart Stroke Prediction using Machine Learning" [8] focuses on the creation of an analytical model based on accessible demographic and health data capable of predicting the occurrence of a stroke. The study explores different classification methods in order to recognize the most important features and risk factors of the disease. According to the study results, the Random Forest algorithm performs better than other classification methods, showing a significant possibility for machine learning usage in this field.

"Unifying Interpretation of Model Predictions" [9] introduces SHAP (SHapley Additive exPlanations), a framework that attempts to unify the interpretation of the predictions made by sophisticated machine learning models, thereby reconciling the conflict between accurate predictions and interpretability. In this paper, it is shown that the proposed framework successfully unifies six previously known methods of attributing importance scores to features into a new family of additive feature importance metrics, which it uniquely satisfies. Additionally, two novel algorithms for estimating the values of these SHAP metrics are provided that demonstrate better computational efficiency and intuitive consistency than previous algorithms.

"Trust me, why should I?: On explaining the predictions made by any classifier" [10] describes a

novel explanation technique for the predictions made by any machine learning classifier called LIME (Local Interpretable Model-Agnostic Explanations). This technique allows one to provide comprehensible and accurate explanations of classifier predictions using locally interpretable models. In combination with techniques of representing representative individual predictions without redundancy, it has been shown that this approach greatly enhances decision-making capacity in critical scenarios involving trust. Such scenarios include trusting particular predictions, comparing classifiers, improving poor classifiers, and not trusting classifiers.

"UCI Machine Learning Repository: Heart Disease Data Set" [11] offers an extensive library of medical information, acting as a basic yardstick for assessing the performance of machine learning classification algorithms within the healthcare industry. It consists of data on patients who have been hospitalized in four different hospitals in the United States, Hungary, and Switzerland, where the original set included 76 different variables. This database proves that using a universally accepted group of 14 medical variables, such as resting blood pressure, serum cholesterol, and maximal heart rate achieved, is immensely successful in creating various prediction models.

4. PROPOSED METHODOLOGIES

The proposed system is an agentic AI-driven heart disease risk prediction application that operates as a full-stack web platform. It is made up of three core layers — a responsive single-page application frontend, a Flask REST API backend, and a multi-agent AI processing engine. The frontend is built entirely with custom HTML, CSS, and JavaScript, styled using a dark glass morphism theme with an interactive 3D particle animation canvas. Users can register or log in, enter their clinical details through three different input modes, view their prediction results with animated visuals, chat with an AI assistant for follow-up questions, download a professionally formatted PDF report, and browse their complete prediction history — all without a single page reload.

The system workflow is as follows:

- (1) The user provides patient information through one of three input methods — a structured form with dropdowns and number fields, a free-text description of symptoms, or by uploading a medical report in PDF format.
- (2) If free text is used, the Helper Agent powered by Llama 3.1 8B through the Groq cloud API converts the natural-language input into a structured JSON object containing the 11 clinical features required by the model.
- (3) If a PDF is uploaded, the backend extracts text using PyPDF2 and parses clinical values through regex-based pattern matching.

- (4) The Prediction Agent loads the trained Random Forest model and the saved feature column structure, cleans and encodes the input data, aligns it with the training layout, and returns both the predicted class and the risk probability percentage.
- (5) The Reasoning Agent powered by Llama 3.3 70B through the Groq API generates a clear, plain-language medical interpretation explaining which health factors contributed to the risk level.
- (6) The Lifestyle Agent produces personalized recommendations covering heart-healthy diet choices, safe exercises, stress reduction techniques, sleep improvement tips, and habits to avoid.
- (7) The Report Agent uses ReportLab to generate a structured PDF report containing user details, a health features summary table, prediction result, risk score, medical reasoning, lifestyle advice, and a safety disclaimer.
- (8) The History Manager saves the complete prediction context as a timestamped JSON entry for future reference.
- (9) The Feedback Agent powers the follow-up chat interface, answering user questions grounded in their specific prediction context while maintaining safe, non-diagnostic language.



Figure 1: Cardiovascular Disease Prediction Pipeline

A. Data Preprocessing:

The input dataset contains a mix of numerical and categorical attributes. Numerical fields such as age, resting blood pressure, cholesterol, maximum heart rate, fasting blood sugar, and Oldpeak are cleaned and cast into their correct numeric types. Categorical fields including sex, chest pain type, resting ECG result, exercise-induced angina, and ST slope are transformed into a machine-readable format using one-hot encoding with the first category dropped to avoid multicollinearity. The target column HeartDisease is then separated from the feature matrix. Once encoding is complete, the resulting column structure is saved to disk as `feature_columns.pkl` using Joblib. This step guarantees that any future user input regardless of whether it arrives through the form, free text, or a PDF upload gets aligned to the exact same feature layout the model was trained on, preventing dimension mismatches at prediction time.

B. Train-Test Split:

The dataset is divided into training and testing subsets using an 80:20 ratio. Stratified sampling is applied during this split to ensure that both classes remain proportionally represented in each subset. A fixed random state of 42 is used throughout, making the entire experiment fully reproducible across different runs and environments.

C. Random Forest Classifier:

Random Forest is an ensemble learning method that constructs a large number of independent decision trees during training and aggregates their individual outputs to arrive at a final prediction. For classification problems, the final class is determined through majority voting across all trees. This ensemble approach makes the model considerably more stable than any single decision tree and naturally reduces the risk of overfitting by averaging out the noise captured by individual trees.

Table I: Random Forest Model Parameters

Parameter	Value
Algorithm	Random Forest Classifier
Number of trees	300
Random state	42
Class weight	Balanced
Train-test split	80:20
Encoding method	One-hot encoding
Model storage	Joblib pickle file

Random Forest was chosen as the final model for several practical reasons. It handles nonlinear relationships between clinical features effectively, works well with mixed data types after encoding, does not require feature scaling, and provides reliable probability estimates through its `predict_proba` method which is critical for computing the risk percentage displayed to the user. Additionally, its balanced class weight configuration ensures the model does not develop bias toward the slightly larger normal class in the dataset.

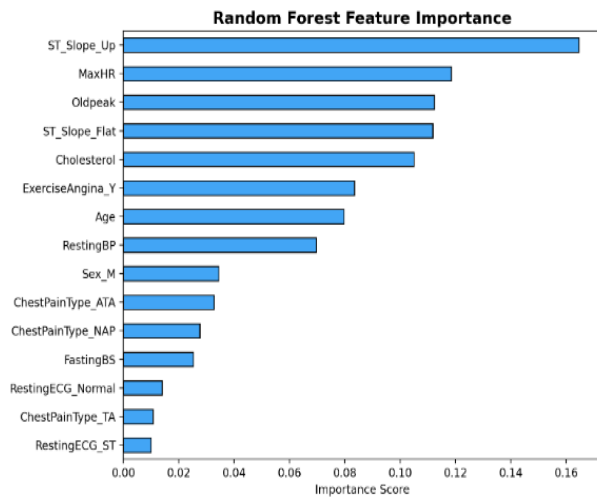


Figure 2. Random Forest Feature Importance

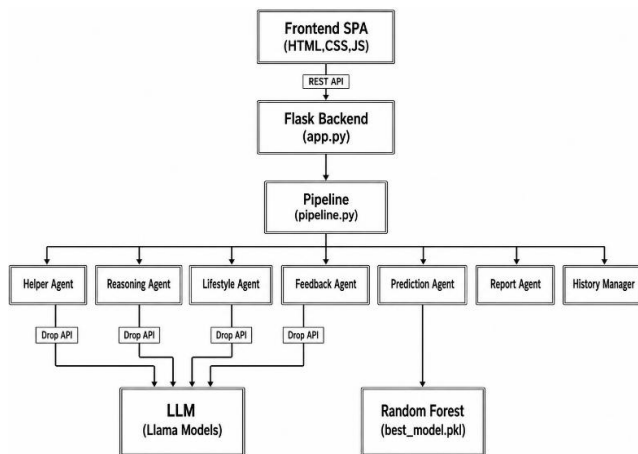


Figure 3. System Architecture — Agentic AI Pipeline

5. IMPLEMENTATION DETAILS

The complete codebase is written in Python and is structured in a way that makes sure that various components of the code are segregated for easier understanding. Training the models is done inside `train_models.py`, where first the heart disease data set is read from a CSV file. The next step is pre-processing it via one-hot encoding followed by splitting the data into training and testing sets. After that comes training the model using the Random Forest classifier with an estimator of 300 and balanced class weights. The model then evaluates itself against multiple performance metrics before saving to disk.

The primary backend system is implemented in `app.py` through the use of Flask, a web development framework. This acts as the REST API interface that provides unique URLs for user sign-up, user sign-in, prediction using forms, prediction using free text, generating PDF reports from uploaded documents, follow-up chat sessions, prediction history lookup, and downloading PDF reports.

Flask provides the frontend in the form of static files. The frontend itself is a custom-designed single-page application developed using simple HTML, CSS, and JavaScript, complete with a dark glass morphism design and an interactive 3D particle canvas animation. The backend is implemented as separate agent files located in the `agents` directory and the pipeline script in the `pipeline.py` file.

Table II: Technologies Used

Component	Technology
Programming Language	Python
Web Framework	Flask REST API
Frontend	HTML, CSS, JavaScript (Custom SPA)
UI Design	Dark Glassmorphism with 3D Particle Canvas
Data Processing	Pandas, NumPy
Machine Learning	Scikit-learn (Random Forest)
AI Reasoning and Recommendations	Groq Cloud API, Llama 3.3 70B, Llama 3.1 8B
PDF Report Generation	ReportLab
PDF Text Extraction	PyPDF2
Data Storage	JSON (<code>history.json</code> , <code>users.json</code>)
Model Persistence	Joblib
Deployment	Render (<code>Procfile</code> , <code>runtime.txt</code>)

6. ML Model Selection

Having preprocessed the dataset, the next key step in the project was to use machine learning to develop predictive models capable of estimating the risk of heart disease. This involved training and testing various classification and ensemble models using the same clinical features in order to select the one that could produce the best possible outcome for our final deployed solution. The different models considered at this stage include Logistic Regression, K-Nearest Neighbors, Random Forest, Extra Trees, Gradient Boosting, XGBoost, and LightGBM.

In essence, the aim of carrying out this comparative study was to evaluate each of these algorithms using conventional metrics such as accuracy, precision, recall, F1 score, and AUC, before choosing one with superior prediction power coupled with reliability. Out of all the algorithms compared here, Random Forest provided the best performance, thus chosen as our prediction algorithm. This algorithm was favored over the others due to its

natural aptitude for identifying nonlinear relations in clinical features, being resilient to overfitting by averaging, and stable in predicting healthcare structured data. Despite the fact that we evaluated KNN and other algorithms as well, none could match the level of performance obtained from Random Forest especially when considering recall and AUC, which are important in a medical screening scenario. Overall, Random Forest will be implemented in the final model of this project where it works with the other components of the agentic AI pipeline to provide risk predictions, probabilities, medical insights, lifestyles, and report downloads.

A. Agent-Based Prediction Pipeline

The agentic architecture takes the clinical input through processing by a number of dedicated agents for various stages of the pipeline:

Helper Agent: Takes the free text input provided by the user describing symptoms and generates the corresponding JSON object that comprises only 11 clinical factors needed for the model. The helper agent is based on the Llama 3.1 8B model provided through the Groq cloud API. The helper agent processes the natural language provided by the user to generate the JSON object according to strict requirements such as the presence of exactly 11 clinical features, with particular attention paid to FastingBS (0 or 1), ExerciseAngina (Y or N), and ChestPainType (ATA, NAP, or TA).

Prediction Agent: Load the pretrained Random Forest classifier model from `best_model.pkl` alongside the saved feature columns from `feature_columns.pkl`. Clean numeric inputs, apply one-hot encoding to categorical values, align the input to the training columns by indexing, and use `predict` and `predict_proba` functions to output a binary class result (low-risk: 0, high-risk: 1) alongside a continuous percentage risk.

Reasoning Agent: Takes the result of the prediction agent, computes the risk percentage, and outputs a human-readable, lay explanation of the medical diagnosis using the full list of features. Using Llama 3.3 70B from the Groq API, the agent finds which health values likely affected the risk, interprets them, highlights risks, and maintains a safe and non-diagnostic language.

Lifestyle Agent: Outputs customized health advice in five main areas: eating healthy for your heart, beginner workouts, managing stress, sleeping better, and unhealthy foods to avoid. Advice is based on the unique patient's feature set and prediction result and powered by Llama 3.1 8B from the Groq API.

Feedback Agent: Powers the post-prediction chat system where users are allowed to ask more questions. Context-aware answer generation occurs, based on the entire prediction context, such as the prediction feature list, risk

score, reasoning, and recommendations regarding the user's lifestyle changes. Runs on the Llama 3.1 8B engine through the Groq API.

Report Agent: Uses the ReportLab package to create a well-formatted PDF document. The report will have a header indicating the title of the document. User details such as name, email, and mobile number will be included at the beginning of the report. There will also be a table listing the user's health features. The report will indicate the prediction output, risk score, medical interpretation of the output, lifestyle recommendations, and disclaimer statement indicating that the report is for education purposes only.

Training Process

Training is done using Heart Failure Prediction data which comprises 918 entries. These include features such as age, gender, type of chest pain, resting blood pressure, cholesterol levels, fast blood sugar, resting ECG, max heart rate achieved, exercise induced angina, oldpeak, and ST slope. The target value is HeartDisease which represents low or high heart disease presence risk. In preprocessing, categorical variables are transformed into numeric values using one-hot encoding with the first column dropped. The encoding schema is saved to `feature_columns.pkl` file so that the same schema could be used at inference time. The data is then split to train and test data sets in a 80/20 ratio using a fixed random seed of 42.

A number of machine learning algorithms were trained and evaluated on this split, and it was found that Random Forests algorithm performed better than others in terms of all the important evaluation criteria. The final configuration of the algorithm involves 300 decision trees, class weights balanced due to class imbalance, and random state as 42. This trained algorithm gave an accuracy of 89.67%, precision of 88.79%, recall of 93.14%, f1-score of 90.91%, and roc-auc of 93.50%. It was pickled using joblib and saved as `best_model.pkl`.

Table III: Reasoning Agent — Clinical Thresholds for Abnormal Feature Detection

Feature	Normal Range	Abnormal Threshold	Clinical Interpretation
Age	< 50 years	≥ 50 years	Elevated baseline cardiovascular risk
Resting BP	90–120 mmHg	> 140 mmHg	Stage 1 hypertension; increases cardiac workload
Cholesterol	< 200 mg/dL	> 240 mg/dL	High LDL promotes plaque formation

Max HR	> 150 bpm	< 100 bpm	Bradycardia or poor exercise capacity
Oldpeak	< 0.5	> 1.0	Significant ST depression indicates ischemia
Exercise Angina	No	Yes	Strong direct predictor of coronary artery disease

7. RESULTS AND EVALUATION

Evaluation of the Trained Random Forest model on the testing dataset was done through the use of five metrics for measuring the classification performance, namely: accuracy, precision, recall, F1-score, and area under ROC curve (ROC-AUC). Five fold-cross validation was used for testing the consistency of the model. These results show that the model is robust enough for healthcare risk screening.

Table IV: Random Forest Model Performance Metrics

Metric	Clinical Significance
Cross-validation F1-score	87.92 %
Accuracy	89.67 %
Precision	88.79 %
Recall	93.14 %
F1-score	90.91%
ROC AUC	93.50%

Prediction Output Examples

To show how the system operates, below are two representative cases from the actual deployment of the application. For each of these cases, the Prediction Agent will calculate the risk score, the Reasoning Agent will formulate the explanation using the Llama 3.3 70B model through the Groq API, and finally, the Lifestyle Agent will generate recommendations using the Llama 3.1 8B model.

High Risk Case (78% chance of heart disease):

Male age of 62 years, whose resting blood pressure is 148 mmHg, cholesterol is 280 mg/dL, fasting blood sugar is 130 mg/dL, max heart rate is 150 beats per minute, angina induced by exercise is present, Oldpeak is 2.3, and he has a flat ST slope. Some of the critical factors found by the Reasoning Agent include: resting blood pressure above normal levels, very high cholesterol level, angina induced

by exercise, and flat ST slope combined with 2.3 mm of ST depression, which suggests that he may have myocardial ischemia. His fasting blood sugar level of 130 mg/dL was interpreted to suggest impaired glucose metabolism.

Case of Low Risk (Risk Probability = 12%):

A 34-year-old woman with resting blood pressure of 118 mmHg, cholesterol of 180 mg/dL, max heart rate of 185 bpm, exercise induced angina not reported, Oldpeak value of 0.0, and upsloping ST segment. The Reasoning Agent found that all of the relevant features lie within the acceptable range, and additionally the young age of 34 makes her even less prone to heart diseases. Lifestyle Agent suggested that she continue the existing healthy lifestyle, with checkups after one-two years.

Comparison With Other Models

In order to prove the choice of Random Forest as our final algorithm, we have trained and tested all seven candidates on the exact same data, using the same 80:20 stratification. The results are presented below.

Table V: Comparative Analysis with Existing Systems

Model	CV F1	Accuracy	Precision	Recall	F1	AUC
Logistic Reg.	86.8 2%	88.59 %	87.16 %	93.1 4%	90.0 5%	93.0 9%
KNN	73.2 7%	69.57 %	72.55 %	72.5 5%	72.5 5%	74.8 6%
Random Forest	87.9 2%	89.67 %	88.79 %	93.1 4%	90.9 1%	93.5 0%
Extra Trees	87.3 5%	89.13 %	87.96 %	93.1 4%	90.4 8%	94.4 8%
Gradient Boost	87.1 5%	87.50 %	89.11 %	88.2 4%	88.6 7%	93.1 9%
XGBoost	86.3 5%	85.87 %	87.25 %	87.2 5%	87.2 5%	92.1 9%
Light GBM	87.0 6%	85.87 %	86.54 %	88.2 4%	87.3 8%	92.3 4%

As is evident from the comparison, Random Forest is the best in terms of accuracy, having 89.67%, and in terms of achieving maximum F1-score with 90.91%. The Extra Trees algorithm was almost equal in several measurements, but managed to get slightly higher AUC measurement – 94.48%, still lower than Random Forest's F1-score. On the contrary, KNN had only 69.57% accuracy and 72.55% F1-score, which means that using such a method in the healthcare domain for predicting a patient's risk is not reasonable since its outcomes were worse. This could be explained by the fact that such a system should use non-linear features.

A high recall rate with 93.14% is also worth mentioning since in this case, the model will help to detect true positive cases (that is those patients who have real risks). As a result, it will prevent false negatives and unnecessary risks for people with cardiovascular issues. It is important to remember that this system is aimed to support clinical decisions rather than substitute them entirely.

Heart Disease Risk Prediction Using Random Forest and Agentic AI

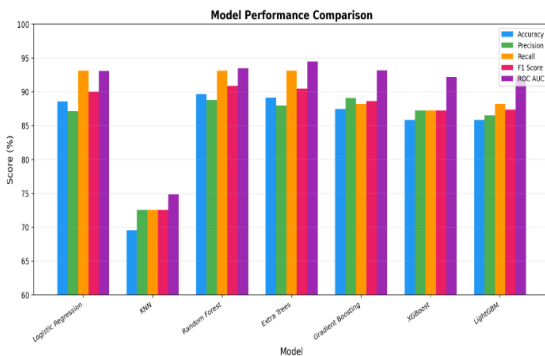


Figure 4: Model Performance Comparison

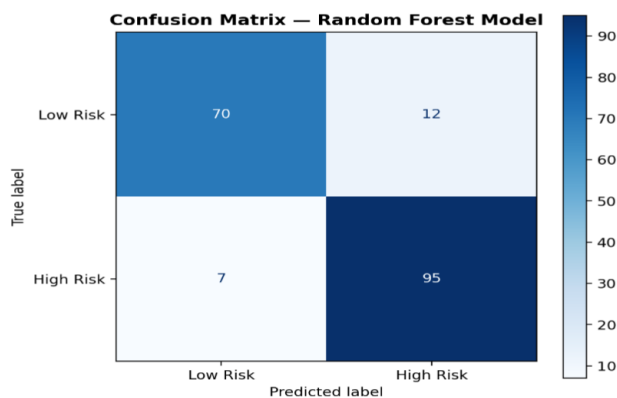


Figure 5: Confusion Matrix — Random Forest Model

8. CONCLUSION

In this study, the full heart disease risk predictor framework consists of an ensemble machine learning algorithm called Random Forest and an agent-based AI design, implemented within the framework of a contemporary web app consisting of a Flask REST API server and a custom single-page application client. The Random Forest classifier, with 300 estimators and class weights balanced, showed outstanding classification accuracy of 89.67 percent, F1-score of 90.91 percent, and ROC-AUC score of 93.50 percent when tested on the test data set. Most importantly, a high recall rate of 93.14 percent is highly significant in a medical screening

application because most individuals at risk can be accurately screened.

What sets this project apart from regular prediction software is the presence of a multi-agent pipeline that consists of multiple agents executing their tasks for each step of the pipeline: from transforming free text or PDF files to extractable structured features, making predictions, using large language models to generate simple medical reasoning, suggesting personal lifestyle advice, facilitating further chat conversations, generating and saving PDF reports, keeping track of prediction history, etc. All of that is provided by means of an attractive glassmorphism frontend interface complete with nice animations and even a 3D particle system.

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